

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEC502 - MICROPROCESSOR AND MICROCONTROLLER

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 If an 8085 instruction has an opcode and two operand bytes, how many machine cycles are required for execution?
 - A. 1
 - B. 2
 - C. 3
 - D. 4
- 2 Which control signal is used by the 8085 to differentiate between memory and I/O operations?
 - A. RD
 - B. WR
 - C. IO/M
 - D. ALE
- 3 While interfacing a 4 KB ROM, how many address lines are required for internal decoding?
 - A. 10
 - B. 11
 - C. 12
 - D. 13
- 4 Which data transfer mode allows data transfer without CPU intervention after initiation?
 - A. Programmed I/O
 - B. Interrupt-driven I/O
 - C. DMA
 - D. Serial I/O

- 5 In memory-mapped I/O, which instruction is used to access I/O devices?
- A. IN
 - B. OUT
 - C. MOV
 - D. SIM
- 6 Which IC is commonly used for D/A conversion in microprocessor applications?
- A. 8255
 - B. DAC0808
 - C. 8259
 - D. ADC0809
- 7 In a stepper motor controller, the rotation angle is controlled by:
- A. Motor speed
 - B. Number of pulses
 - C. Supply voltage
 - D. Coil resistance
- 8 Which register in 8051 is used to configure timer modes?
- A. TMOD
 - B. TCON
 - C. IE
 - D. SCON
- 9 How many interrupt sources are available in the standard 8051 microcontroller?
- A. 3
 - B. 4
 - C. 5
 - D. 6
- 10 Which instruction is used to return from an interrupt service routine in 8051?
- A. RET
 - B. RETI
 - C. SJMP
 - D. AJMP
- 11 Which register controls the serial communication mode in 8051?
- A. IE
 - B. TMOD
 - C. SCON
 - D. PCON
- 12 Which peripheral IC is used for programmable parallel I/O interfacing?
- A. 8251
 - B. 8255
 - C. 8259
 - D. 8237

Q2. Solve the following.

- A) Draw and explain the functional block diagram of 8085 microprocessor and describe the role of ALU, timing and control unit, and registers during instruction execution. 6

- B) An 8085 executes a three-byte instruction. Explain the fetch and execution cycle with the help of timing signals and machine cycles. 6
- Q3. Solve the following.
- A) Design a memory interfacing scheme to connect a 4 KB ROM and 4 KB RAM with 8085 using address decoding. Draw the memory map. 6
- B) Explain the DMA mode of data transfer with a neat block diagram. How does DMA improve system performance compared to programmed I/O? 6
- Q4. Solve Any Two of the following.
- A) Explain the interfacing of an ADC with 8085 and describe the steps involved in analog-to-digital conversion. 6
- B) Describe the interfacing of a multiplexed seven-segment LED display with a microprocessor. Explain how display refreshing is achieved. 6
- C) Explain the working of a stepper motor controller using a microprocessor. How is speed and direction controlled? 6
- Q5. Solve Any Two of the following.
- A) Explain the 8051 architecture with reference to memory organization, special function registers, and I/O ports. 6
- B) Describe the timer/counter operation in 8051. Explain any one timer mode with its application. 6
- C) Explain the interrupt structure of 8051 microcontroller and describe the interrupt handling sequence. 6
- Q6. Solve Any Two of the following.
- A) Write and explain an 8051 assembly language program to generate a square wave using a timer. 6
- B) Explain the serial communication in 8051 using RS-232. Describe the role of SCON and TMOD registers. 6
- C) Explain the interfacing of ADC/DAC with a microcontroller and mention its applications. 6

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